



AI Infrastructure Architecture Evolution

2015-2026

Centriq · Oryon · Cloud AI · Dragonfly
HBC · Connectivity · Edge AI · Systems · Software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Scope and executive synthesis

Families and weights: deep on Centriq/Oryon/Dragonfly CPUs and Cloud AI/Dragonfly accelerators; full on datacenter connectivity; proportional on client, edge and RAN derivatives.

Start: 2015, including the first server effort and the mobile technology baseline.

Cutoff: 28 September 2026; current disclosures include Investor Day and AI Infra Summit 2026.

Focus: CPU lineage, inference dataflow, memory capacity/bandwidth, fabric and rack integration.

Exclusions: exhaustive handset/modem SKUs; no merchant datacenter training GPU or separately specified Qualcomm AI NIC is established by the reviewed portfolio.

Qualcomm's infrastructure history is not a continuous server-CPU succession. Centriq was a real commercial Arm server processor, but the next major CPU story comes through Nuvia-derived Oryon client products and the future Dragonfly C1000. In parallel, Cloud AI 100 establishes a dedicated inference architecture, Ultra expands model capacity through a multi-SoC card, and AI200/AI250/AI300 move the product boundary to the rack. Connectivity assets from Alphawave and software from Modular complete parts of that broader platform.

The dominant architectural bet is inference efficiency rather than reproducing a general-purpose training GPU. Large local capacity can reduce model partitioning, and near-memory execution can reduce the cost of moving weights during decode. These benefits are conditional on dataflow, model placement and software. AI250's effective HBC bandwidth must therefore remain separate from conventional external-memory bandwidth. The platform's long-term CPU and accelerator roadmap extends beyond the cutoff: C1000 production is planned for the second half of 2028, not a shipping 2026 server.

02 Units, product boundaries and roadmap status

A die, card, server and rack are different accounting scopes. Cloud AI 100 Ultra contains four SoCs; its 128 GB memory and 576 MB SRAM are card totals. AI200's 43 TB is a rounded rack capacity, whereas 768 GB is per card. Rack power is not card TBP. Bandwidth is decimal GB/s or TB/s, and link direction is stated explicitly. PCIe headline bandwidth is a raw interface budget before protocol overhead. Ethernet Gb/s must be divided by eight before comparison with memory GB/s.

The standard accelerator table preserves dense BF16, FP8, FP6/FP4 and FP64 columns. Where the reviewed evidence supplies FP16 or INT8 instead, those values are reported separately rather than relabelled. n/d means not established; n/a means not applicable. A product announcement, customer sampling, a demonstration and general availability are different milestones. In particular, AI200's 2025 launch forecast of 2026 availability is followed by a 2026 investor roadmap labelled sampling FY26; the reviewed September demo does not alone establish broad GA.

03 CPU generation atlas

CPU and related client references · 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Codename	GA	Status	Core µarch	Cores / threads
Centriq 2460	Falkor	2017-11	Historical shipping	Custom Armv8	48 / 48
Snapdragon X Elite	Oryon	2024	Historical shipping	Oryon 1	12 / 12
Snapdragon X2 Elite Extreme	X2E-96-100	2026 systems	Shipping / available	Oryon 3	18 / 18
Dragonfly C1000	n/d	2028 H2 target	Roadmap	Oryon server	250+ / n/d

CPU and related client references · 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
Centriq 2460	Samsung 10 nm	Monolithic; 398 mm2; 18B transistors	Shared by 2-core duplex; capacity n/d	60 MB
Snapdragon X Elite	4 nm	Client SoC; 3 CPU clusters	12 MB / 4-core cluster	System-cache accounting differs
Snapdragon X2 Elite Extreme	3 nm	Client SoC; Prime + Performance	n/d	53 MB total CPU cache; not all L3
Dragonfly C1000	n/d	Chiplet architecture	n/d	n/d

CPU and related client references · 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
Centriq 2460	DDR4; 6; 2667 MT/s; ~128 GB/s	1S; QSB internal	32 PCIe 3; CXL n/a	120 W	NEON; AArch64
Snapdragon X Elite	LPDDR5X; 135 GB/s	Client SoC; n/a	PCIe; CXL n/a	OEM configurable; n/d	Arm64; NEON
Snapdragon X2 Elite Extreme	LPDDR5X; 192-bit; 228 GB/s	Client SoC; n/a	PCIe; CXL n/a	OEM configurable; n/d	Arm64; details n/d
Dragonfly C1000	LPDDR; channel count/rate n/d	n/d	PCIe 7; CXL; lanes n/d	n/d	Server ISA details n/d

Client entries explain the core-development lineage; their memory, power and I/O are not server specifications. C1000 is not a socketed Snapdragon X2 Elite. Nor does the acquisition of Ventana make C1000 a RISC-V processor: Qualcomm explicitly identifies Oryon server cores for C1000, while Ventana adds a separate RISC-V capability. No second commercial Centriq generation is established here.

04 Centriq and Falkor: the first server architecture

Centriq 2400 began sampling in 2016 and commercially shipped in November 2017. Falkor is a custom, single-threaded, AArch64-only core. Hot Chips describes an issue path accepting three instructions plus a direct branch and dispatch of up to eight internal operations. A two-core duplex shares L2 and a QSB interface. The segmented bidirectional coherent ring uses shortest-path routing; its reported aggregate bandwidth above 250 GB/s is an internal-fabric figure, not six-channel DDR payload bandwidth.

The duplex L2 uses 128-byte cache lines, ECC and an inclusive relationship to L1 data caches. The disclosed minimum L2 hit latency is fifteen cycles. Distributed L3 includes allocation controls for quality of service, an important feature when many independent tenants share a socket. Six DDR4 channels and 32 PCIe 3 lanes define a balanced scale-out server of its time, but offer a much smaller external accelerator budget than later AI hosts. At 48 cores, approximately 128 GB/s of channel bandwidth means about 2.67 GB/s per core before contention and protocol effects.

The launch disclosure identifies Samsung 10 nm, 18 billion transistors and a 398 mm² monolithic die. This is implementation evidence even though a matching full Centriq ISSCC paper was not established in the reviewed sources. OCP 2017 supplies the system connection: Qualcomm and Microsoft demonstrated a Centriq-based Open Compute motherboard. That collaboration is evidence of platform development, not proof of Azure public-instance GA. The later break in the server roadmap means that commercial continuity cannot be inferred from the technical quality of Falkor.

05 Oryon: client silicon rebuilds the CPU lineage

The 2021 Nuvia acquisition leads to the Oryon CPU introduced in Snapdragon X Elite and commercial PCs in 2024. Hot Chips 2024 and its IEEE Micro follow-up provide the architectural anchor. Three four-core clusters combine substantial near-core cache with a wide out-of-order machine. The design emphasizes instruction supply, branch recovery and outstanding memory operations rather than width alone. The core's client implementation demonstrates engineering capability; it does not demonstrate server RAS, multi-socket coherence or datacenter memory capacity.

Qualcomm's Hot Chips 2024 slides give the first Oryon core a 192 KB instruction cache, a 96 KB data cache, eight-instruction decode and eight-micro-op retirement. Six integer and four 128-bit vector execution pipes feed a wide out-of-order machine. The load/store system supports up to four operations per cycle, with a 192-entry load queue and 56-entry store queue. Four cores share a 12 MB L2 at CPU frequency; three clusters connect through a fabric with 6 MB system-level cache to a 135 GB/s LPDDR5X interface. These layers explain why the advertised 42 MB cache total is neither a per-core cache nor a unified last-level cache.

Oryon then crosses into mobile and the third-generation core returns to Snapdragon X2 Elite. The X2E-96-100 reference combines twelve Prime cores with six Performance cores, 53 MB total CPU cache and 228 GB/s LPDDR5X bandwidth. Its 80 TOPS Hexagon NPU is a separate engine, not CPU vector throughput. The mixed core organization and OEM power envelope are client choices. For the server report, the important continuity is custom core design, cache and power-management expertise, and a common software ISA; memory channels and host links must be re-established for C1000.

ISSCC/JSSC-related circuit work explains another part of Qualcomm's efficiency heritage. Keith Bowman's adaptive-clock research senses supply droop and adjusts clock behavior to avoid paying a permanent worst-case timing margin. The documented Snapdragon 820 connection makes that a legitimate example of research reaching a product. It does not establish that Centriq, Oryon or C1000 uses an identical circuit. The ISSCC 2025 edge-generative-AI tutorial is useful system context, not a transistor-level disclosure of a Dragonfly accelerator.

06 Dragonfly C1000 and the RISC-V branch

Investor Day 2026 announces a chiplet-based C1000 with more than 250 custom Oryon server cores, a core-frequency target above 5 GHz, PCIe 7/CXL connectivity and an LPDDR memory subsystem. Optional HBC attachment is part of the disclosed direction. The architectural shift from Centriq is substantial: a larger package-level design and a CPU positioned around agent orchestration, general cloud services and accelerator head-node work. The Meta agreement places first-generation production in the second half of 2028 and anticipates subsequent generations.

The headline I/O claim of more than 2 TB/s is not normalized here to a lane count or one-way payload figure, because the disclosed direction and aggregation basis are insufficient. Cache sizes, memory-channel count, process partitioning, socket coherence and SKU TDP remain material disclosure gaps. LPDDR can reduce memory energy, but capacity, serviceability, packaging and board topology must be evaluated together. An announced frequency and a core-count ceiling are not evidence that every core sustains that clock at the eventual package power limit.

Qualcomm acquired Ventana in December 2025, adding RISC-V CPU expertise alongside Oryon. Ventana's earlier Veyron compute-chiplet work is relevant as an acquired design capability, but the reviewed Qualcomm disclosure does not define a shipping successor SKU or transfer its specifications to C1000. The two instruction-set paths should therefore remain separate in a portfolio map. Their potential common ground is physical design, chiplet integration, verification and server software engineering, rather than binary compatibility.

07 Accelerator generation atlas

Inference accelerators in the common comparison schema · 1/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
Cloud AI 100 Standard	AIC	2021 systems	Historical shipping	7 nm	1 SoC per card
Cloud AI 100 Pro	AIC	2021 systems	Historical shipping	7 nm	1 SoC per card
Cloud AI 100 Ultra	4 x AIC SoC	2023-2024	Shipping / available	7 nm	4 SoCs + PCIe switch
Dragonfly AI200	Inference ASIC	GA n/d	Sampling / demonstrated	n/d	n/d
Dragonfly AI250	HBC Gen 1	2027 target	Roadmap	n/d	Near-memory architecture; die count n/d
Dragonfly AI300	HBC Gen 2	FY28 roadmap	Roadmap	n/d	n/d

Inference accelerators in the common comparison schema · 2/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
Cloud AI 100 Standard	SKU-enabled count n/d	n/d	n/d	n/d	n/a
Cloud AI 100 Pro	16 AIC nominal	n/d	n/d	n/d	n/a
Cloud AI 100 Ultra	4 x 16 AIC nominal	n/d	n/d	n/d	n/a
Dragonfly AI200	n/d	n/d	n/d	n/d	n/d
Dragonfly AI250	n/d	n/d	n/d	n/d	n/d
Dragonfly AI300	n/d	n/d	n/d	n/d	n/d

Inference accelerators in the common comparison schema · 3/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM	Scale-up: links; GB/s per direction; domain
Cloud AI 100 Standard	LPDDR4X; n/a; 16; 0.137	126 MB SRAM	PCIe; no dedicated scale-up link
Cloud AI 100 Pro	LPDDR4X; n/a; 32; 0.137	144 MB SRAM	PCIe; topology-dependent
Cloud AI 100 Ultra	LPDDR4X; n/a; 128; 0.548	576 MB SRAM	On-card PCIe switching
Dragonfly AI200	LPDDR5X; n/a; 768/card; ~7.39 derived	n/d	PCIe 6; link count/BW n/d
Dragonfly AI250	HBC; n/a; 768/card; 133 effective	n/d	PCIe 6; link count/BW n/d
Dragonfly AI300	HBC; capacity n/d; 54x AI200 effective BW target	n/d	UALink / ESUN roadmap; all-to-all

Inference accelerators in the common comparison schema · 4/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Host link	Power / cooling	Form factor
Cloud AI 100 Standard	PCIe 4 x8	75 W	HHHL PCIe
Cloud AI 100 Pro	PCIe 4 x8	75 W	HHHL PCIe
Cloud AI 100 Ultra	PCIe 4 x16	150 W	FH 3/4 length PCIe
Dragonfly AI200	PCIe; details n/d	Card n/d; rack 140 kW	Card / ORv3 rack
Dragonfly AI250	n/d	Card n/d; air/DLC rack	Card / ORv3 rack
Dragonfly AI300	n/d	n/d; liquid-cooled rack	Rack platform

Published legacy-card compute figures

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Card	FP16 TFLOP/s	INT8 TOPS	Rated power	Source basis
Standard	110	325	75 W	SDK 1.12 SKU table
Pro	125	375	75 W	SDK 1.12 SKU table
Ultra	288 product / 290 SDK	870	150 W	Small document revision/rounding difference retained

08 Cloud AI 100: programmable inference dataflow

The 2019 announcement is followed by selected-customer shipments in 2020 and commercial system activity in 2021. Hot Chips 2021 presents the inference architecture. Each AI core separates tensor, vector and scalar work. Tensor units handle the dense linear algebra; vector units process elementwise and other non-matrix operations; a multithreaded VLIW scalar processor coordinates execution. This is not a GPU warp machine. The compiler must expose parallelism, schedule movement and keep specialized engines occupied across a model graph.

The SDK describes 8 MB vector tightly coupled memory and 1 MB shared L2 within each nominal AI core, producing 144 MB of on-chip storage across sixteen cores. Separate compute, memory and configuration networks serve different traffic classes. External LPDDR4X provides capacity; local SRAM provides reuse and predictable scheduling. DMA and tiling are therefore first-class architectural mechanisms. A model that reuses weights or activations locally can amortize external traffic, while a large decode workload repeatedly reading weights can become memory-bound despite substantial tensor capacity.

Standard, Pro and the early DM.2 variants address different power and deployment envelopes. The SoC's architectural peak, an initial launch target and a shipping card's rated clock point are not interchangeable. For this reason the report uses the explicit Standard/Pro SKU table for card-level comparisons. Host PCIe 4 x8 provides about 15.75 GB/s per direction before protocol overhead, far below the card's 137 GB/s memory figure. Keeping model state resident on the card is consequently much more attractive than repeatedly streaming it across the host link.

09 Cloud AI 100 Ultra: capacity by replication

Ultra combines four Cloud AI 100 SoCs and a PCIe switch on a 150 W card. Each SoC has its own local memory; the card totals 128 GB LPDDR4X, 576 MB SRAM and 548 GB/s of summed memory bandwidth. The host interface expands to PCIe 4 x16, while each SoC attaches to the switch over x8. Four chips on one board do not create one uniform-latency shared SRAM or one unconstrained 548 GB/s path for every operation. Model partitioning and communication scheduling remain necessary.

The 2023 product is explicitly positioned around generative inference. Its capacity allows some large quantized models to remain on one card, reducing the number of boards and host/network crossings. That is a model-placement advantage, not evidence of higher training throughput. Weight precision, KV-cache size, context length and batch size determine whether a model fits. Independent 2025 NRP measurements confirm that the platform can be used for real LLM serving, but their energy ratios depend on the selected models, comparison configurations and utilization; they are not portable chip-level efficiency constants.

10 AI200: the inference rack becomes the product

AI200 moves from a low-power PCIe-card story to a rack-scale inference platform. The published configuration contains 56 cards, 768 GB LPDDR per card, approximately 43 TB per rack and 0.414 PB/s aggregate memory bandwidth. It specifies PCIe 6 scale-up and Ethernet/RoCE scale-out in an ORv3 rack, with a 140 kW liquid-cooled configuration. Dividing rack memory bandwidth by 56 gives approximately 7.39 TB/s per card as an arithmetic average. It does not disclose a single die's memory interface or prove equal bandwidth for every placement.

A September 2026 demonstration places Kimi-K2.5 on a single AI200 card. That is relevant evidence for model-capacity and software progress, but the demo does not establish sustained throughput, response-time distribution, model precision or broad customer availability. Architecturally, the large capacity can reduce weight sharding, while PCIe scale-up and RoCE still have to move activations, expert-routing traffic and collectives. Model residency and communication efficiency solve different constraints.

11 AI250 and AI300: near-memory execution

AI250 introduces High Bandwidth Compute Gen 1, with a published 133 TB/s effective memory bandwidth per card and 7.4 PB/s per rack. Qualcomm describes this as eighteen times AI200's effective bandwidth. The key mechanism is moving computation closer to memory, so a useful computation need not pay the same external data movement as a conventional processor reading every operand over a memory bus. Effective bandwidth is a workload/architecture equivalence metric; treating it as an ordinary HBM pin-rate specification would produce a misleading roofline comparison.

The natural target is memory-sensitive decode and long-context serving. Prefill may remain more compute-intensive, so disaggregating the phases can improve resource matching but adds transfer and scheduling costs. HBC's practical value depends on which operators execute near memory, the precision they support, locality across memory partitions, and how intermediate results reach the rest of the machine. The public investor material outlines the approach but does not disclose enough circuit, stack or inter-die detail to reconstruct a complete implementation.

AI300 extends the roadmap to HBC Gen 2, with UALink/ESUN scale-up, copper and optical scale-out, and a single-hop all-to-all rack objective. The investor roadmap quotes a fifty-four-times AI200 effective-bandwidth target. This is a future architecture claim, not measured shipping throughput. C1000 is shown as the future CPU partner. Per-card link rates, radix, packet semantics, memory capacity and complete power limits remain undisclosed in the reviewed material, so the report does not manufacture a rack bisection figure.

12 Connectivity: from Alphawave assets to Dragonfly DSPs

The Alphawave transaction closed on 18 December 2025, earlier than the original 2026 expectation. It adds high-speed wired connectivity and custom-silicon capabilities, which matter because a rack platform needs more than compute engines. The 2026 portfolio spans electrical SerDes, die-to-die interfaces, optical DSPs, active electrical cable retimers and custom silicon. A protocol controller, a PHY, a retimer, an optical module and a complete NIC occupy different layers; they must not be collapsed into a single AI-networking product count.

Dragonfly O100/O200 are PAM4 optical DSP families; CO400 is a coherent-lite 16QAM optical DSP. CU100/CU200 address PAM4 active electrical cables. CU100 is described as a 4 nm retimer. The 100G/200G/400G labels identify signaling/product classes, not a fully specified NIC port count. Investor Day distinguishes volume-production 800G connectivity, 1.6T scaling in the 2026-2027 period and later 3.2T development. A 448G SerDes roadmap is likewise a technology direction, not evidence of a shipping 448G-per-lane rack.

Network and connectivity product classes · 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	GA / milestone	Status	Class	Ports × speed	SerDes
Dragonfly O100	2026 portfolio	Announced; GA n/d	Optical DSP	n/a	100G PAM4 class
Dragonfly O200	2026 portfolio	Announced; GA n/d	Optical DSP	n/a	200G PAM4 class
Dragonfly CO400	2026 portfolio	Announced; GA n/d	Coherent-lite DSP	n/a	400G 16QAM
Dragonfly CU100	2026 portfolio	Announced; GA n/d	AEC DSP; 4 nm	n/a	100G PAM4
Dragonfly CU200	2026 portfolio	Announced; GA n/d	AEC DSP	n/a	200G PAM4
Dragonwing X100	2021 disclosure	Shipping / available	5G RAN inline accelerator	n/d	n/d

Network and connectivity product classes · 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Host interface	Programmable engine	Embedded CPU	On-card memory
Dragonfly O100	n/a	Signal processing; packet pipeline n/a	n/a	n/a
Dragonfly O200	n/a	DSP	n/a	n/a
Dragonfly CO400	n/a	DSP	n/a	n/a
Dragonfly CU100	n/a	DSP	n/a	n/a
Dragonfly CU200	n/a	DSP	n/a	n/a
Dragonwing X100	PCIe	5G L1 software	n/d	n/d

Network and connectivity product classes · 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Transport / RDMA	Congestion control	Key offloads	Power
Dragonfly O100	n/a	n/a	Optical link DSP	n/d
Dragonfly O200	n/a	n/a	Optical link DSP	n/d
Dragonfly CO400	n/a	n/a	Longer-reach optics	n/d
Dragonfly CU100	n/a	n/a	Electrical retiming	n/d
Dragonfly CU200	n/a	n/a	Electrical retiming	n/d
Dragonwing X100	O-RAN fronthaul; AI RDMA n/a	n/a	Baseband L1 processing	n/d

13 Interconnect and system topology

Normalized interconnect boundaries · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
QSB	2017	n/a	n/a	n/d; >250 GB/s aggregate internal
Cloud AI 100 host	2021	16 GT/s	8	~15.75
Ultra host / local	2023	16 GT/s	16 host; 8/SoC	~31.5 host; ~15.75/SoC
AI200 / AI250 scale-up	2026-2027	64 GT/s	n/d	n/d
AI300 UALink / ESUN	FY28	n/d	n/d	n/d

Normalized interconnect boundaries · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
QSB	n/a	Segmented coherent ring	48 cores	CPU/cache/I/O coherence
Cloud AI 100 host	1	PCIe	1 card	DMA / host I/O
Ultra host / local	4 SoC links	On-card PCIe switch	4 SoCs	Partitioned local memory; PCIe movement
AI200 / AI250 scale-up	n/d	PCIe 6	Rack solution; topology n/d	Coherence not established
AI300 UALink / ESUN	n/d	Single-hop all-to-all target	n/d	Protocol details n/d

System reference envelopes · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
Centriq OCP motherboard	2017	Centriq 2400	1 socket
Ultra inference server	2023-2025	Host CPU + PCIe cards	4 SoCs per card; system-dependent
AI200 ORv3	2026	56 cards	PCIe 6 rack integration
AI250 ORv3	2027 target	43 TB rack; card count n/d here	PCIe 6
AI300 + C1000	2028 roadmap	Future CPU + HBC accelerators	All-to-all target

System reference envelopes · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
Centriq OCP motherboard	n/a	n/d	Server-dependent
Ultra inference server	n/d	n/d	Air, platform-dependent
AI200 ORv3	RoCE; per-card rate n/d	140 kW	DLC reference; air option disclosed
AI250 ORv3	RoCE; rate n/d	n/d	Air / DLC
AI300 + C1000	n/d	n/d	Liquid

The rack references include NICs and switches, but the reviewed product pages do not specify a complete Qualcomm-branded AI NIC with transport, congestion-control and offload details. It is therefore more accurate to show RoCE as the system transport and the DSPs as connectivity components. Retiming a signal does not implement RDMA, congestion control or collective acceleration. Similarly, UALink participation and controller IP do not by themselves establish a merchant scale-up switch SKU.

14 Adreno, Hexagon, Dragonwing and RAN

Qualcomm's GPU line is Adreno, integrated into Snapdragon and related SoCs. Its role includes graphics and flexible local compute; it is not the architecture identity of Cloud AI 100. Hexagon combines DSP/vector/tensor capabilities for edge AI, with newer products emphasizing dedicated NPU execution. Snapdragon X and mobile Oryon products show CPU, GPU and NPU sharing a constrained memory and thermal envelope. Peak values across these engines cannot simply be added as one usable model throughput because operators, formats and data movement differ.

Dragonwing is the industrial/embedded and infrastructure brand; Dragonfly is the datacenter portfolio. The Cloud AI 100 Ultra can also serve on-premises inference appliances, connecting the edge and datacenter stories without turning every embedded product into a server accelerator. X100 is especially easy to misread: Dragonwing X100 is a 5G RAN inline accelerator for Layer 1 baseband processing, whereas Cloud AI 100 is an AI inference family. MWC disclosures belong to telecom infrastructure, not evidence of an AI training fabric.

15 Software and academic evidence

Cloud AI SDK separates model compilation from runtime execution and exposes hardware-aware model preparation. The future Dragonfly stack adds the AI Inference Suite, infrastructure management, AIC ahead-of-time compilation, QCCL, NPU-direct and orchestration integration. Modular's acquisition completed in July 2026, adding a software platform whose existing cross-hardware capability is distinct from future Dragonfly optimization. An acquisition announcement or a stack diagram does not establish that every operator and numerical format is production-ready on every accelerator generation.

ASPLOS 2025's Fast On-device LLM Inference with NPUs is a useful implementation-facing study: it partitions variable-length prompts, handles quantization outliers across engines and schedules blocks according to CPU/GPU/NPU affinity. It studies how to use available hardware, not an undisclosed Dragonfly microarchitecture. ISCA 2023 shared-microexponent work and the OCP microscaling paper explain low-precision representation choices, but neither alone proves native support in Cloud AI 100 silicon. Academic sponsorship, a program-committee affiliation or a paper using a Snapdragon phone is not a product disclosure.

Software timeline and enablement boundary

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Release / component	Date	Hardware enabled	Key features
Cloud AI SDK	2021-2026	Cloud AI 100 / Ultra	Compile, deploy, profile; model support is release-specific
AI Engine Direct / QNN	2022-2026	Snapdragon / Hexagon / Adreno	On-device heterogeneous execution
AI Inference Suite / AIMS	2026 disclosure	Dragonfly	Serving, fleet management, QCCL and NPU-direct roadmap
Modular acquisition	2026-07	Cross-hardware AI software	Completed acquisition; future hardware enablement separate

16 Quantitative balance and deployment trade-offs

Derived ratios with explicit scope

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Configuration	Derived quantity	Value	Interpretation
Centriq 48C	DRAM/core; LLC/core	~2.67 GB/s; 1.25 MB	Theoretical equal share
AI 100 Pro	Memory BW / FP16 peak	0.001096 byte/FLOP	137 GB/s / 125 TFLOP/s; not BF16
AI 100 Ultra	Memory BW / FP16 peak	0.001903 byte/FLOP	548 GB/s / 288 TFLOP/s product figure
AI 100 Pro	Host / local memory bandwidth	~0.115	15.75 / 137; one direction; raw host link
AI 100 Ultra	Host / summed local memory BW	~0.0575	31.5 / 548; four memory domains
AI200	Memory/card; BW/card average	768 GB; ~7.39 TB/s	414 TB/s / 56 cards; no card-power inference
AI 100 Pro	FP16 peak / rated TDP	1.67 TFLOP/s/W	125 / 75; not measured tokens/W
AI 100 Ultra	FP16 peak / rated TDP	1.92 TFLOP/s/W	288 / 150; not whole-system energy

These ratios expose the portfolio’s trade-offs. Ultra quadruples card memory capacity relative to Pro while doubling rated card power, but only doubles the host-link width; residency becomes more important. AI200 increases the unit of integration and cooling burden. AI250 attacks decode movement with a different memory-compute organization. Dense BF16/FP8 bytes-per-FLOP, HBM ratios, scale-out bandwidth per card and measured rack tokens/W remain n/d where the required normalized inputs are absent. Substituting marketing multipliers would hide the architectural question rather than answer it.

17 Conference-to-product index

Verified disclosure anchors · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Venue	Year	Paper / talk / disclosure	Presenter / authors
Hot Chips	2017	Qualcomm Centriq 2400 Processor / Falkor	Qualcomm
Hot Chips	2021	Cloud AI 100: Scalable, High Performance and Low Latency Deep Learning Inference Accelerator	Qualcomm
Hot Chips / IEEE Micro	2024 / 2025	Snapdragon X Elite Qualcomm Oryon CPU: Design & Architecture Overview	Gerard Williams / Qualcomm
ISSCC / JSSC lineage	2015 / 2016	Auto-calibrating adaptive clock distribution	Keith Bowman et al.
ISSCC	2025	Generative AI on Edge Devices: Models, Hardware and Systems	Paul Whatmough
ASPLOS	2025	Fast On-device LLM Inference with NPUs	Daliang Xu et al.
ISCA	2023	With Shared Microexponents, A Little Shifting Goes a Long Way	B. Rouhani et al.
OCP Summit	2017	Centriq Open Compute motherboard with Microsoft	Qualcomm / Microsoft
MWC-era announcement	2021	5G DU X100 accelerator	Qualcomm
Snapdragon Summit	2023 / 2025	X Elite / X2 Elite	Qualcomm
Investor Day	2026	Dragonfly CPU, HBC, accelerators and connectivity	Tony Pialis
AI Infra Summit	2026	Dragonfly inference demonstrations	Qualcomm

Verified disclosure anchors · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Venue	Product	Evidence type	Source
Hot Chips	Centriq	Architecture	C01 E01
Hot Chips	Cloud AI 100	Architecture	C06 P01
Hot Chips / IEEE Micro	Oryon	Architecture	C07 C08 C02
ISSCC / JSSC lineage	Circuit research; Snapdragon 820 link	Circuit research	C04
ISSCC	Edge AI context	Tutorial	C03
ASPLOS	Snapdragon-class NPU use	System research	R04
ISCA	Numerical-format background	Related research	R02
OCP Summit	Centriq	System demonstration	E12
MWC-era announcement	X100 RAN	Product / system	E11
Snapdragon Summit	Oryon client lineage	Product launch	E17 P12
Investor Day	C1000 / AI200-300 / DSPs	Roadmap / system	P08 E09 E10
AI Infra Summit	AI200	Demonstration	E07

MICRO and HPCA were checked as architecture-research venues, but the reviewed record does not establish a primary Centriq, Cloud AI or Dragonfly product-architecture paper there. DAC and VLSI can illuminate implementation methods; OFC and Hot Interconnects are relevant to the acquired connectivity stack; Computex and company events establish product milestones. AWS re:Invent, Microsoft Ignite, Google Cloud Next and GTC require a specific partner deployment to count as evidence. The report does not turn generic ecosystem participation into a product presentation or claim exhaustive conference attendance.

18 Codename and product crosswalk

Disambiguation map

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Name	Product identity	Boundary
Falkor	Custom CPU core in Centriq 2400	Not Neoverse; not Oryon
Oryon	Custom CPU family; client and future server	Client SoC != C1000
Cloud AI 100 / Ultra	Inference ASIC and PCIe cards	Not Adreno GPU; not X100 RAN
Dragonfly	Datacenter CPU, accelerator and connectivity brand	C1000 / AI200 / AI250 / AI300 / DSPs
Dragonwing X100	5G baseband inline accelerator	Telecom L1 offload
HBC	High Bandwidth Compute	Near-memory compute; not HBM synonym

19 Integrated stack across four eras

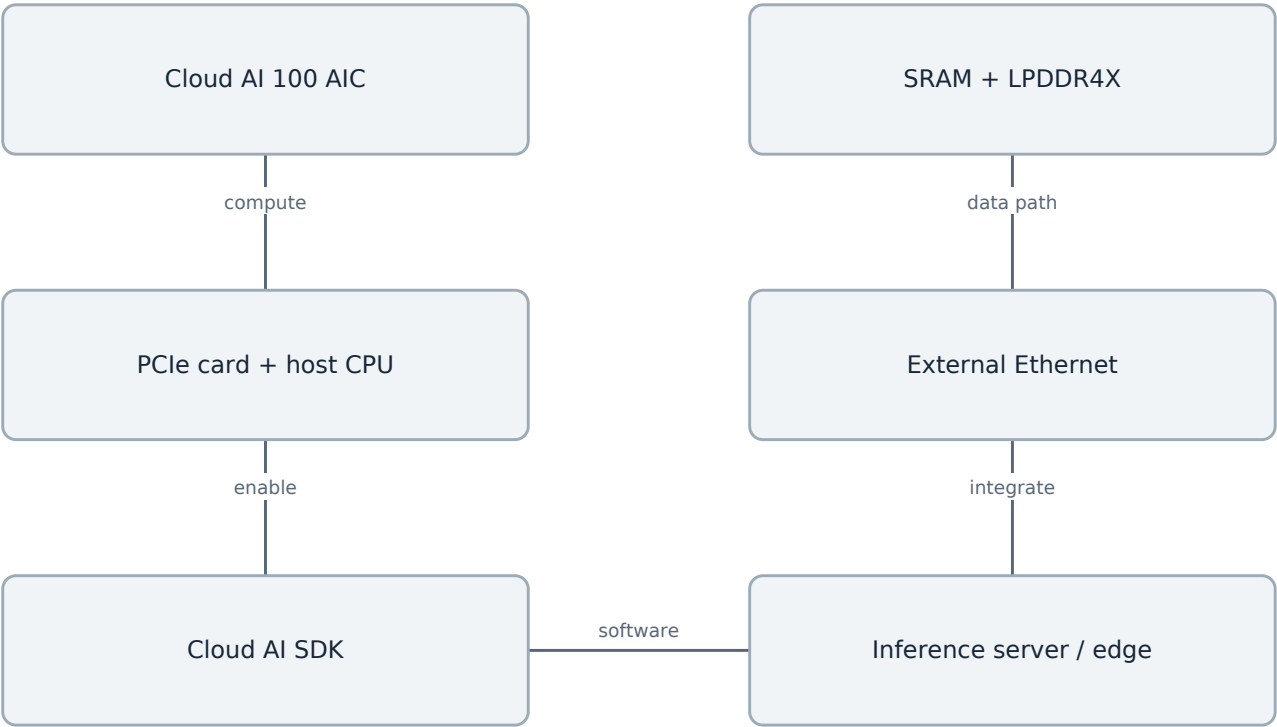
Connections show architectural composition. A roadmap block remains a roadmap in the system view; placing two future products in one figure does not establish a shipping platform.

2015-2018: the scale-out server attempt



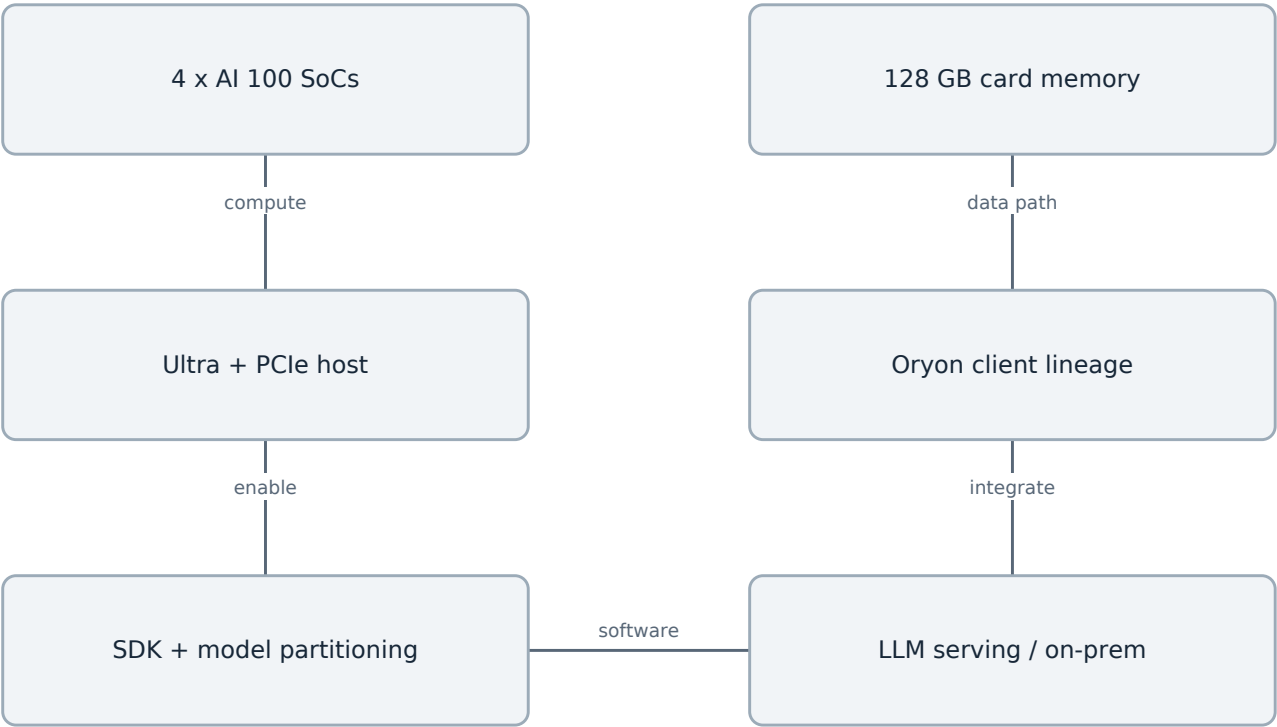
Conceptual composition, with statuses defined in the product chapters; external NIC ownership is not inferred.

2019-2022: inference cards



Conceptual composition, with statuses defined in the product chapters; external NIC ownership is not inferred.

2023-2025: model capacity and new CPU cores



Conceptual composition, with statuses defined in the product chapters; external NIC ownership is not inferred.

2026 disclosures: the future rack portfolio



Conceptual composition, with statuses defined in the product chapters; external NIC ownership is not inferred.

Bandwidth hierarchy by era

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Era / reference	On-die / die-to-die	Local memory	CPU / scale-up	Scale-out
2017 Centriq	QSB >250 GB/s aggregate	~128 GB/s DDR4	PCIe 3; ~15.75 GB/s per x16 direction	External NIC; n/d
2021 AI 100 Pro	3 NoCs; aggregate scope not normalized	137 GB/s	PCIe 4 x8 ~15.75 GB/s/direction	External NIC; n/d
2023 Ultra	4 discrete SoCs; PCIe switch	548 GB/s summed	PCIe 4 x16 ~31.5 GB/s/direction	External NIC; n/d
2026 Dragonfly disclosure	HBC implementation details n/d	AI200 ~7.39 TB/s/card; AI250 133 effective	PCIe 6 / future UALink; n/d	RoCE; per-card bandwidth n/d

20 Engineering conclusions

Qualcomm's strongest continuity is in energy-conscious programmable compute and data movement, not an uninterrupted server product cadence. Centriq contributes a real server baseline; Oryon rebuilds custom CPU capability; Cloud AI 100 provides a distinct inference engine; Ultra demonstrates capacity scaling through board-level replication. Dragonfly's next step is harder: coordinating CPU, near-memory acceleration, high-speed connectivity, software and cooling at rack scale. Its success should be assessed with model fit, latency-constrained token throughput, communication overhead and measured system power, while keeping future production targets separate from present capabilities.

Source map

- 01 Scope and executive synthesis - C01, E02, E05, E07, E08, E10, E14, P01, P05, P08
- 02 Units, product boundaries and roadmap status - E06, E07, P01, P04, P08
- 03 CPU generation atlas - C01, C02, C07, C08, E01, E02, E10, E13, E16, P07, P08, P12
- 04 Centriq and Falkor: the first server architecture - C01, E01, E02, E12
- 05 Oryon: client silicon rebuilds the CPU lineage - C02, C03, C04, C05, C08, E15, P12
- 06 Dragonfly C1000 and the RISC-V branch - E10, E13, P07, P08
- 07 Accelerator generation atlas - E04, E05, E06, E07, P01, P02, P03, P04, P05, P06, P08
- 08 Cloud AI 100: programmable inference dataflow - C06, E03, E04, P01, P02
- 09 Cloud AI 100 Ultra: capacity by replication - E05, P01, P03, R01
- 10 AI200: the inference rack becomes the product - E07, P04
- 11 AI250 and AI300: near-memory execution - P05, P06, P08
- 12 Connectivity: from Alphawave assets to Dragonfly DSPs - E08, E11, P08, P09, P10, P11
- 13 Interconnect and system topology - C01, E10, E12, P01, P04, P05, P06, P08, P09, R01
- 14 Adreno, Hexagon, Dragonwing and RAN - E11, P03, P11, P12, P14, R04
- 15 Software and academic evidence - E14, P01, P08, P13, P14, R02, R03, R04
- 16 Quantitative balance and deployment trade-offs - C01, E02, P01, P03, P04, P05
- 17 Conference-to-product index - C01, C02, C03, C04, C06, C07, C08, E01, E07, E09, E10, E11, E12, E17, P01, P08, P12, R02, R04
- 18 Codename and product crosswalk - C01, C07, P01, P07, P08, P09, P11
- 19 Integrated stack across four eras - C01, E10, P01, P04, P05, P06, P08
- 20 Engineering conclusions - C01, E10, E14, P01, P08

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